

Practice Exam Set A — Solutions

Part I: Generative Models

Exercise A — VAE

Answer: A

With $z' = 1$, $\mathbf{x}' = [2, -1]^\top$, $q(z|\mathbf{x}') = \mathcal{N}(z | x_1 + x_2, 1) = \mathcal{N}(z | 1, 1)$:

Reconstruction term: The likelihood mean is $[z', -z']^\top = [1, -1]^\top$.

$$\ln p(\mathbf{x}' | z') = \ln \mathcal{N}([2, -1]^\top | [1, -1]^\top, \mathbf{I}_2) = -\frac{1}{2}[(2-1)^2 + (-1+1)^2] - \ln(2\pi) = -\frac{1}{2} - \ln(2\pi).$$

KL term ($\mu_q = 1$, $\sigma_q^2 = 1$):

$$\text{KL}[\mathcal{N}(1, 1) \| \mathcal{N}(0, 1)] = \ln \frac{1}{1} + \frac{1 + (1-0)^2}{2 \cdot 1} - \frac{1}{2} = 0 + 1 - \frac{1}{2} = \frac{1}{2}.$$

$$\text{ELBO}(\mathbf{x}') = -\frac{1}{2} - \ln(2\pi) - \frac{1}{2} = -1 - \ln(2\pi).$$

Exercise B — Normalizing Flows

Answer: B

We need $T^{-1}([1, 1]^\top)$. Using the given table: $T_2([1, 0]^\top) = [1, 1]^\top$, so $T_2^{-1}([1, 1]^\top) = [1, 0]^\top$. Then $T_1([0, 0]^\top) = [1, 0]^\top$, so $T_1^{-1}([1, 0]^\top) = [0, 0]^\top$. Hence $\mathbf{u} = [0, 0]^\top$.

Jacobian determinant:

$$\det J_T([0, 0]^\top) = \det J_{T_2}(T_1([0, 0]^\top)) \cdot \det J_{T_1}([0, 0]^\top) = \det J_{T_2}([1, 0]^\top) \cdot 2 = 3 \cdot 2 = 6.$$

Base density: $p_{\mathbf{u}}([0, 0]^\top) = \frac{1}{2\pi} \exp(-\frac{1}{2}(0+0)) = \frac{1}{2\pi}$.

$$p_{\mathbf{x}}([1, 1]^\top) = \frac{p_{\mathbf{u}}([0, 0]^\top)}{|\det J_T|} = \frac{1/2\pi}{6} = \frac{1}{12\pi}.$$

Exercise C — DDPM

Answer: A

Compute $\bar{\alpha}_2 = \prod_{t=1}^2 (1 - \beta_t) = (1 - \frac{1}{2})(1 - \frac{1}{2}) = \frac{1}{4}$.

$$\sqrt{\bar{\alpha}_2} \mathbf{x}_0 = \frac{1}{2}[4, -4]^\top = [2, -2]^\top, \quad 1 - \bar{\alpha}_2 = \frac{3}{4}.$$

$$q(\mathbf{x}_2 | \mathbf{x}_0 = [4, -4]^\top) = \mathcal{N}(\mathbf{x}_2 | [2, -2]^\top, \frac{3}{4}\mathbf{I}).$$

Part II: Representation Learning

Exercise D — Pull-back metric

Answer: A

For $f(\mathbf{x}) = W_2 \text{ReLU}(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$, the Jacobian is

$$J_{\mathbf{x}} = W_2 \underbrace{\text{diag}(\mathbf{1}[W_1 \mathbf{x} + \mathbf{b}_1 > 0])}_{\Lambda_1} W_1.$$

Hence $G_{\mathbf{x}} = J_{\mathbf{x}}^{\top} J_{\mathbf{x}} = (W_2 \Lambda_1 W_1)^{\top} (W_2 \Lambda_1 W_1) = W_1^{\top} \Lambda_1 W_2^{\top} W_2 \Lambda_1 W_1$.

The key distinction: Λ_1 uses $W_1 \mathbf{x} + \mathbf{b}_1$ (with bias), not $W_1 \mathbf{x}$ (without bias).

Exercise E — Curve energy

E.1: For $f(\mathbf{x}) = [x_1^2, x_1 x_2, x_2]^{\top}$:

$$J_{\mathbf{x}} = \begin{bmatrix} 2x_1 & 0 \\ x_2 & x_1 \\ 0 & 1 \end{bmatrix}, \quad G_{\mathbf{x}} = J_{\mathbf{x}}^{\top} J_{\mathbf{x}} = \begin{bmatrix} 4x_1^2 + x_2^2 & x_1 x_2 \\ x_1 x_2 & x_1^2 + 1 \end{bmatrix}.$$

E.2: For $\mathbf{c}_1(t) = t[1, 0]^{\top}$, $\dot{\mathbf{c}}_1 = [1, 0]^{\top}$, $G_{\mathbf{c}_1(t)} = \begin{bmatrix} 4t^2 & 0 \\ 0 & t^2 + 1 \end{bmatrix}$:

$$\dot{\mathbf{c}}_1^{\top} G \dot{\mathbf{c}}_1 = 4t^2, \quad E(\mathbf{c}_1) = \int_0^1 4t^2 dt = \left[\frac{4t^3}{3} \right]_0^1 = \boxed{\frac{4}{3}}.$$

For $\mathbf{c}_2(t) = t[0, 1]^{\top}$, $\dot{\mathbf{c}}_2 = [0, 1]^{\top}$, $G_{\mathbf{c}_2(t)} = \begin{bmatrix} t^2 & 0 \\ 0 & 1 \end{bmatrix}$:

$$\dot{\mathbf{c}}_2^{\top} G \dot{\mathbf{c}}_2 = 1, \quad E(\mathbf{c}_2) = \int_0^1 1 dt = \boxed{1}.$$

Exercise F — Riemannian integration

F.1: $G_{\mathbf{x}} = \text{diag}(x_2^2, x_1^2)$, so $\det G_{\mathbf{x}} = x_1^2 x_2^2$, $\sqrt{\det G_{\mathbf{x}}} = x_1 x_2$ on $[0, 1]^2$.

$$\boxed{dM(\mathbf{x}) = x_1 x_2 dx_1 dx_2.}$$

F.2:

$$\begin{aligned} I_1 &= \int_0^1 \int_0^1 x_1^2 \cdot x_1 x_2 dx_1 dx_2 = \int_0^1 x_1^3 dx_1 \cdot \int_0^1 x_2 dx_2 = \frac{1}{4} \cdot \frac{1}{2} = \boxed{\frac{1}{8}}. \\ I_2 &= \int_0^1 \int_0^1 x_2 \cdot x_1 x_2 dx_1 dx_2 = \int_0^1 x_1 dx_1 \cdot \int_0^1 x_2^2 dx_2 = \frac{1}{2} \cdot \frac{1}{3} = \boxed{\frac{1}{6}}. \end{aligned}$$

Exercise G — ELBO and evidence

Starting from the definition:

$$\begin{aligned} \text{ELBO}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \left[\ln \frac{p(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} \right] = \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}|\mathbf{x}) p(\mathbf{x})}{q(\mathbf{z}|\mathbf{x})} \right] && [\text{factorize } p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z}|\mathbf{x}) p(\mathbf{x})] \\ &= \ln p(\mathbf{x}) + \mathbb{E}_q \left[\ln \frac{p(\mathbf{z}|\mathbf{x})}{q(\mathbf{z}|\mathbf{x})} \right] && [\ln p(\mathbf{x}) \text{ constant w.r.t. } \mathbf{z}] \\ &= \ln p(\mathbf{x}) - \text{KL}[q(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}|\mathbf{x})]. \quad \square \end{aligned}$$

Part III: Graph Models

Exercise H — Graph Laplacians

H.1: From L_{rw} , row a : $[1, -1, 0, 0, 0]$ means node a has degree 1 with sole neighbour b . Row b : $[-\frac{1}{2}, 1, -\frac{1}{2}, 0, 0]$ means b has degree 2, neighbours a and c . Similarly $c-d$, $d-e$. The graph is a **path** $a-b-c-d-e$.

H.2: Transition matrix $T = I - L_{\text{rw}} = D^{-1}A$:

$$T_{ab} = 1; \quad T_{ba} = T_{bc} = \frac{1}{2}; \quad T_{cb} = T_{cd} = \frac{1}{2}; \quad \text{etc.}$$

Starting at a : step $1 \rightarrow b$ (prob 1). Step 2: $b \rightarrow a$ (prob $\frac{1}{2}$) or $b \rightarrow c$ (prob $\frac{1}{2}$). Step 3:

$$P(\text{at } b \text{ after } 3) = P(\text{at } a \text{ at step } 2) \cdot 1 + P(\text{at } c \text{ at step } 2) \cdot \frac{1}{2} = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{3}{4}}.$$

Exercise I — Weisfeiler-Lehman test

Graph: triangle $a-b-c-a$ plus pendant $a-d$. Degrees: $d_a = 3$, $d_b = d_c = 2$, $d_d = 1$.

I.1:

- **A TRUE:** $\ell_a^{(0)} = d_a = 3$.
- **B FALSE:** $\ell_d^{(0)} = d_d = 1 \neq 2$.
- **C TRUE:** $\mathcal{N}(b) = \{a, c\}$ and $\mathcal{N}(c) = \{a, b\}$ give multisets $\{3, 2\}$ and $\{3, 2\}$, so $\ell_b^{(1)} = \ell_c^{(1)} = \text{hash}(\{3, 2\})$.
- **D FALSE:** $\ell_a^{(1)} = \text{hash}(\{2, 2, 1\}) \neq \text{hash}(\{3, 2\}) = \ell_b^{(1)}$.
- **E FALSE:** $\ell_d^{(1)} = \text{hash}(\{3\}) \neq \text{hash}(\{3, 2\}) = \ell_b^{(1)}$.

True: A, C

I.2: Round 0 partition: $\{\{a\}, \{b, c\}, \{d\}\}$. After round 1: a gets $\text{hash}(\{2, 2, 1\})$, b and c both get $\text{hash}(\{3, 2\})$, d gets $\text{hash}(\{3\})$ — same partition $\{\{a\}, \{b, c\}, \{d\}\}$. No new distinctions were made, so the algorithm halts.

$$\boxed{\text{Number of rounds} = 1.}$$

Exercise J — Shallow embeddings

Answer: A

$$\|\mathbf{z}_u - \mathbf{z}_v\|^2 = 0.25 \text{ with } \mathbf{z}_u = [0, 1]^\top:$$

$$(x - 0)^2 + (y - 1)^2 = 0.25 = (0.5)^2.$$

This is a circle of radius 0.5 centred at $\mathbf{z}_u = [0, 1]^\top$, matching option A.

Exercise K — Message passing**K.1:** Star graph: $\mathcal{N}(a) = \{b, c, d\}$, $\mathcal{N}(b) = \mathcal{N}(c) = \mathcal{N}(d) = \{a\}$.

$$h_a^{(1)} = 1 + 1 \cdot 1 + 1 \cdot (1 + 1 + 1) = 1 + 1 + 3 = 5,$$

$$h_b^{(1)} = h_c^{(1)} = h_d^{(1)} = 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$$h_a^{(2)} = 1 + 1 \cdot 5 + 1 \cdot (3 + 3 + 3) = 1 + 5 + 9 = \boxed{15},$$

$$h_b^{(2)} = h_c^{(2)} = h_d^{(2)} = 1 + 1 \cdot 3 + 1 \cdot 5 = \boxed{9}.$$

K.2: $M = w_1 I + w_2 A = I + A$ where A (ordering a, b, c, d) has $A_{ab} = A_{ac} = A_{ad} = A_{ba} = A_{ca} = A_{da} = 1$, rest 0:

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$